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Who Holds Sovereign Debt and Why It Matters

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1 Big picture

- **Question.** Does *who* holds sovereign debt affect sovereign borrowing costs and funding risk?
- **Setting.** The paper constructs a new sovereign-debt-holdings data set for 101 countries from 1990 to 2018, decomposing debt into domestic and foreign investors, and within each location into banks, nonbanks, and official creditors.
- **Core challenge.** Simple holding shares do not tell us whether investor composition matters for pricing. Composition matters only if different investor groups have different demand elasticities for sovereign debt.
- **Main reduced-form result.** Private nonbank investors absorb a disproportionate share of sovereign debt expansions. Across all countries, they take up 62% of new debt on the margin even though they hold only 44% on average.
- **Main structural result.** In emerging market (EM) sovereign debt, nonbank investors—especially **foreign nonbanks**—have the highest demand elasticity with respect to expected excess returns.
- **Main counterfactual result.** For the average EM borrower, a 10% increase in debt raises yields by about 5.8%. Without nonbank investors, the same debt increase raises yields by about 8.4%.
- **Economic message.** Sovereigns are not just exposed to the *level* of debt demand, but to the *composition* of their investor base. EM sovereigns in particular are vulnerable to losing nonbank investors.
- **Why the paper matters.** It brings the demand-system approach to asset pricing into sovereign debt markets and shows that investor heterogeneity is central for sovereign funding risk and debt sustainability.

2 Data and measurement (key objects)

2.1 Observed sovereign-by-year panel

- **Unit of observation.** Sovereign issuer n in year t .
- **Coverage.** 1947 country-year observations for 101 countries over 1990–2018.
- **Total sovereign debt.** Total general government debt is built from the IMF Historical Public Debt Database (HPDD) and Global Debt Database (GDD), multiplied by World Bank GDP.
- **Foreign total holdings.** Constructed using the methodology of Avdjiev et al., combining International Investment Position (IIP), Quarterly External Debt Statistics (QEDS), and BIS international banking and debt securities statistics.
- **Domestic total holdings.** Defined residually as total debt minus foreign-held debt.
- **Investor-group decomposition.** The six main holding groups are:
 - Domestic banks (DomBK),
 - Domestic nonbanks (DomNB),

- Domestic central bank / official (DomCB),
 - Foreign banks (ForBK),
 - Foreign nonbanks (ForNB),
 - Foreign official creditors (ForOff).
- **Construction details.** Foreign official holdings for AEs and China come from Arslanalp and Tsuda; for other countries they are bilateral and multilateral official lending from the World Bank Debtor Reporting System. Domestic bank and central bank holdings come from IMF IFS and central-bank websites. Domestic and foreign nonbank positions are residuals after subtracting bank and official holdings.
 - **Country groups.** For many exercises, countries are grouped into advanced economies (AEs), emerging markets (EMs), and developing countries (DCs).

2.2 Disaggregated follow-up datasets

- **U.S. Treasury TIC data.** Used to decompose domestic nonbanks into money market funds, households/hedge funds, insurance and pension funds, other financial institutions, and nonfinancial corporations.
- **U.K. Gilt holder data.** Used to decompose domestic nonbanks into funds/other financial institutions, insurance and pensions, and nonfinancial corporations.
- **Euro Area Securities Holdings Statistics (SHS).** Used to decompose euro-area foreign nonbank investors into insurance and pensions, other financial institutions (largely investment funds), households, and nonfinancial corporations.

2.3 Key measured objects

- **Book value of sovereign debt:**

$$D(n) = \sum_{i=1}^I H_i(n).$$

- **Market value of holdings:**

$$H_{i,t}^m(n) \equiv P_t(n) H_{i,t}(n).$$

- **Market value of debt:**

$$D_t^m(n) \equiv D_t(n) P_t(n).$$

- **Holding share of investor group i :**

$$\theta_i(n) \equiv \frac{H_i^m(n)}{D^m(n)}.$$

- **Marginal share of investor group i :**

$$a_i(n) \equiv \frac{\partial H_i(n)}{\partial D(n)}.$$

- **Country characteristics.** The baseline vector is

$$x_t(n) = \{\log GDP, GDP\ growth, inflation, \log(Exports/GDP), sovereign\ ratings\}.$$

- **Borrowing cost.** The 5-year sovereign bond yield, mostly in local currency for EM debt.

2.4 Unobservables and timing

- **Latent demand.** Each investor group has an unobserved demand component for each country's debt.
- **Timing.** The main dataset is annual, so the analysis is medium-run rather than high-frequency.
- **Important limitation.** The data do not contain the maturity structure or currency composition of debt, the exact location of foreign investors, or investors' nonsovereign portfolios.
- **Implication.** The paper can study issuer-side composition and demand elasticities, but not a full investor-side global portfolio problem for all sovereigns.

3 Models and empirical specifications (all equations + interpretation)

3.1 Accounting decomposition of sovereign debt

The basic accounting identity is

$$D(n) = \sum_{i=1}^I H_i(n). \quad (1)$$

- **Interpretation.** This is purely an accounting decomposition: total debt equals the sum of holdings across all investor groups.
- **Why it matters.** The whole paper starts from the fact that the debt of every sovereign is fully allocated across creditor types.

3.2 Reduced-form marginal holder regression

To measure who absorbs additional debt on the margin, the paper estimates

$$\frac{H_{i,t}(n) - H_{i,t-1}(n)}{D_{t-1}(n)} = a_i \frac{D_t(n) - D_{t-1}(n)}{D_{t-1}(n)} + a(n) + a_t + \varepsilon_{i,t}(n), \quad \forall i. \quad (2)$$

- **Interpretation.** a_i is the marginal share of new sovereign debt taken up by investor group i .
- **Restriction.** Because the holdings must add up to total debt,

$$\sum_{i=1}^I a_i = 1.$$

- **Important warning.** This is a reduced-form decomposition, not a causal demand estimate.

3.3 Single-country and euro-area decompositions

For the U.S. and U.K. granular datasets, the same decomposition is run country by country:

$$\frac{H_{i,t}(n) - H_{i,t-1}(n)}{D_{t-1}(n)} = a_0(n) + a_i(n) \frac{D_t(n) - D_{t-1}(n)}{D_{t-1}(n)} + e_{i,t}(n), \quad n \in \{US, UK\}. \quad (3)$$

For the euro-area SHS, the paper decomposes the marginal response of euro-area nonbanks:

$$\frac{H_{j,t}^{EA}(n) - H_{j,t-1}^{EA}(n)}{H_{NB,t-1}^{EA}(n)} = a_j \frac{H_{NB,t}^{EA}(n) - H_{NB,t-1}^{EA}(n)}{H_{NB,t-1}^{EA}(n)} + a^{EA}(n) + a_t^{EA} + \varepsilon_{j,t}^{EA}(n), \quad \forall j. \quad (4)$$

- **Interpretation.** Equation (3) asks who holds marginal U.S. or U.K. issuance; Equation (4) asks which types of euro-area nonbanks drive the euro-area nonbank response.

3.4 Sovereign debt pricing framework: supply and market clearing

Debt supply is summarized as

$$D_t(n) = D(P_t(n), x_t(n), D_{t-1}(n)),$$

where price, fundamentals, and lagged debt all affect issuance.

Define the market value of investor group i 's holdings as $H_{i,t}^m(n) = P_t(n)H_{i,t}(n)$. Then the market-clearing condition is

$$P_t(n)D_t(n) = \sum_{i=1}^I H_{i,t}^m(n). \quad (5)$$

- **Interpretation.** The equilibrium price of sovereign debt is jointly determined by debt supply and investor demand.
- **Core insight.** Investor composition matters only if different investor groups have different demand functions.

3.5 Investor demand system for EM sovereign debt

The starting demand specification is in portfolio-share form:

$$\delta_{i,t}(n) \equiv \ln\left(\frac{\omega_{i,t}(n)}{\omega_{i,t}(0)}\right) = \tilde{\lambda}_{0,i}\mu_{i,t}(n) + \tilde{\lambda}'_{1,i}x_t(n) + \tilde{\lambda}_{i,t} + \tilde{\lambda}_i(n) + \tilde{\varepsilon}_{i,t}(n), \quad (6)$$

where $\omega_{i,t}(n)$ is the portfolio share in country n 's sovereign debt, $\omega_{i,t}(0)$ is the outside asset share, and $\mu_{i,t}(n)$ is investor group i 's expected excess return.

Because the data do not observe all portfolio shares directly, the paper rewrites the demand system in holdings form:

$$\ln H_{i,t}^m(n) = \lambda_{0,i}\mu_{i,t}(n) + \lambda'_{1,i}x_t(n) + \lambda_{i,t} + \lambda_i(n) + \varepsilon_{i,t}(n). \quad (7)$$

- **Interpretation.** Holdings of sovereign debt rise when the expected excess return is higher or when country characteristics are more attractive.

- **Crucial assumptions for EM estimation.**

- Foreign investors in EM debt are outside the EM issuer sample.
- Foreign investors are U.S.-dollar-based.
- Domestic investors are local-currency-based.
- For domestic investors, the outside asset is proxied by GDP because country-specific domestic portfolios are not directly observed.

3.6 Expected excess return forecasting equations

For foreign investors, the paper predicts expected excess returns in dollars using

$$r_{t+1}(n) + s_{t+1}(n) - s_t(n) - y_t(US) = \phi^f p_t(n) + \phi^r (s_t(n) - z_t(n)) + \chi^f(n) + \nu_{t+1}^f(n), \quad (8)$$

where $r_{t+1}(n)$ is the local-currency holding return, $s_t(n)$ is the log dollar price of local currency, and $z_t(n)$ is the price-level differential entering the real exchange rate.

For domestic investors, expected excess returns are local-currency returns:

$$r_{t+1}(n) - y_t(n) = \phi^d p_t(n) + \chi^d(n) + \nu_{t+1}^d(n). \quad (9)$$

- **Interpretation.** Foreign investors care about bond returns *and* exchange-rate movements; domestic investors care only about local-currency excess returns.

3.7 Instrument construction from issuer-side market clearing

Because expected excess returns depend on endogenous prices, the paper constructs an instrument in three steps.

First, estimate holdings using characteristics only:

$$\ln H_{i,t}^m(n) = \beta'_{1,i} x_t(n) + \beta_{i,t} + \beta_i(n) + u_{i,t}(n). \quad (10)$$

Second, estimate debt supply using lagged debt and characteristics:

$$\ln d_t(n) = \gamma_0 + \gamma_1 \ln d_{t-1}(n) + \gamma_2 x_t(n) + \gamma(n) + \gamma_t + v_{i,t}(n), \quad (11)$$

where $d_t(n) = D_t(n)/Y_t(n)$ is debt-to-GDP.

Third, plug the fitted values into market clearing:

$$\hat{P}_t(n) \hat{d}_t(n) Y_t(n) = \sum_{i=1}^I \hat{H}_{i,t}^m(n). \quad (12)$$

- **Interpretation.** The hypothetical price $\hat{P}_t(n)$ is driven only by exogenous characteristics and lagged debt, so it can instrument for endogenous expected excess returns.
- **Why this is novel.** The instrument comes from the issuer side because the data fully allocate each sovereign's debt across investor groups.

3.8 Demand elasticities

The implied price elasticity of investor group i is

$$\eta_i \equiv \frac{\partial \ln H_{i,t}^m(n)}{\partial \ln P_t(n)} = \lambda_{0,i}(1 - \omega_i(n))\phi_i, \quad (13)$$

where $\phi_i = \phi^d$ for domestic investors and $\phi_i = \phi^f$ for foreign investors.

The elasticity with respect to characteristic $x_k(n)$ is

$$\psi_{i,k} \equiv \frac{\partial \ln H_{i,t}^m(n)}{\partial x_k(n)} = \lambda_{1,i,k}(1 - \omega_i(n)). \quad (14)$$

- **Interpretation.** η_i tells us how responsive each investor group is to price or return changes; $\psi_{i,k}$ tells us how sensitive demand is to country fundamentals.
- **Main empirical result.** Foreign nonbanks have the largest elasticity, followed by foreign banks and domestic nonbanks.

3.9 Financing-cost counterfactuals

Using average holding shares, the yield response to an increase in the market value of debt is

$$\frac{\partial y(n)}{\partial \ln D^m(n)} = -\frac{1}{T} \left(\sum_{i=1}^I \theta_i(n) \eta_i \right)^{-1}. \quad (15)$$

Using marginal shares instead, the alternative expression is

$$\frac{\partial y(n)}{\partial \ln D^m(n)} = -\frac{1}{T} \frac{\sum_{i=1}^I a_i(n)/(\eta_i - 1)}{1 + \sum_{i=1}^I a_i(n)/(\eta_i - 1)}. \quad (16)$$

The paper then links marginal shares to average shares and elasticities through Equation (17): more elastic investors with larger average holdings absorb more new issuance on the margin.

For characteristic changes, the financing sensitivity is

$$\frac{\partial y(n)}{\partial x_k(n)} = -\frac{1}{T} \frac{\sum_{i=1}^I \psi_{i,k}(n) \theta_i(n)}{1 - \sum_{i=1}^I \theta_i(n) \eta_i}. \quad (18)$$

For latent demand shocks by investor group j ,

$$\frac{\partial y(n)}{\partial \varepsilon_j(n)} = -\frac{1}{T} \frac{\theta_j(n)}{1 - \sum_{i=1}^I \theta_i(n) \eta_i}. \quad (19)$$

- **Interpretation.** Borrowing-cost sensitivity depends on two things:
 - who holds the debt (the θ_i 's), and
 - how elastic those investors are (the η_i 's).
- **Big economic implication.** Losing high-elasticity investors—especially nonbanks—makes sovereign funding much more fragile.

3.10 Advanced-economy robustness exercise

For AEs, the paper cannot estimate the same full demand system because foreign investors are often other AEs inside the sample. As a robustness exercise, it estimates:

$$\ln H_{i,t}^m(n) = \alpha_{i,0} \ln y_t(n) + \alpha'_{i,1} x_t(n) + \alpha_i(n) + \alpha_{i,t} + u_{i,t}(n). \quad (20)$$

- **Interpretation.** This is not a full structural AE demand system, but it still checks whether the same broad pattern appears in AE holdings.
- **Result.** Foreign nonbanks remain the most yield-responsive AE investor group.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- The marginal-holder regressions in Section 1 are informative but not causal. They only describe who absorbs debt expansions, not why.
- Prices and expected returns are endogenous in sovereign debt markets because they are equilibrium outcomes of supply and demand.
- Therefore, regressing holdings directly on yields or returns would confound investor demand with latent demand shocks and sovereign borrowing conditions.

4.2 What identifies the EM demand system

- The identifying advantage of the dataset is that it fully allocates each sovereign's debt across investor groups.
- This full market-clearing identity allows the paper to construct an issuer-side instrument for expected excess returns using predicted supply and predicted holdings from observables alone.
- The approach is closest to Koijen and Yogo's demand-system method, but adapted to a setting where the researcher observes the *full creditor composition* of each issuer rather than detailed investor portfolios.

4.3 Why the EM sample is the natural structural application

- The assumption that foreign investors are outside the estimation sample is plausible for EM debt because many foreign EM investors are advanced-economy investors.
- The assumption that foreign investors are dollar-based is also more plausible for EM debt than for AE debt.
- These assumptions are not credible enough for AE sovereigns, which is why the paper does not run the same full structural demand estimation for the AE sample.

4.4 Why the nonbank-composition evidence is informative

- The main aggregate result could in principle hide heterogeneity inside the nonbank category.
- The paper therefore uses three disaggregated datasets—U.S. Treasuries, U.K. Gilts, and euro-area SHS—to show that the dominant component of nonbanks is investment-fund-like institutions.
- This matters because the aggregate counterfactuals should be interpreted largely as statements about the presence or absence of investment-fund demand.

4.5 What pins down the counterfactuals

- The demand coefficients come from the IV estimation of the holdings equations.
- The elasticities require portfolio-share proxies for each investor group:
 - domestic banks and nonbanks from IMF financial soundness indicators,
 - foreign nonbanks from SHS,
 - foreign banks from BIS cross-border claims.
- The financing-cost counterfactuals then combine these elasticities with either average holding shares or marginal holding shares.

4.6 Relation to the literature

- The paper extends the demand-system approach to sovereign debt markets.
- It complements the sovereign-bank nexus literature by comparing banks, nonbanks, and official creditors jointly.
- It also connects to work on reserve demand, official lending, and safe-asset demand, but adds a creditor-composition angle centered on sovereign funding risk.

5 Tables and figures

5.1 Figure 1: Trends in general government debt

Read:

- Government debt-to-GDP rises strongly over time for both AEs and EM/DCs.
- The domestic share of holdings falls in AEs but rises in EMs.
- Nonbank shares fluctuate over time, suggesting that the marginal importance of nonbanks may vary with market conditions.
- The figure motivates the paper’s central question: when debt rises, who actually absorbs it?

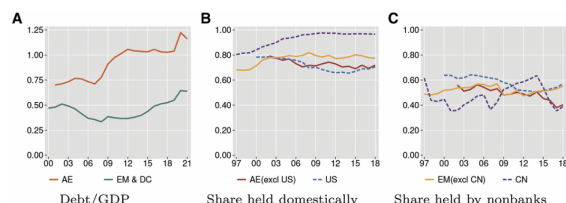


Figure 1
Trends in general government debt
Panel A plots the time series of general government debt-to-GDP ratio for the advanced economies (AE) and for the emerging market and developing countries (EM & DC). Data are from IMF WEO. Panel B plots the share of general government debt held by domestic investors, aggregated by country group. Panel C plots the share of general government debt held by nonbanks. Panels B and C consist of a balanced sample from our data set of 17 AEs including the United States (US) and 22 EMs including China (CN).

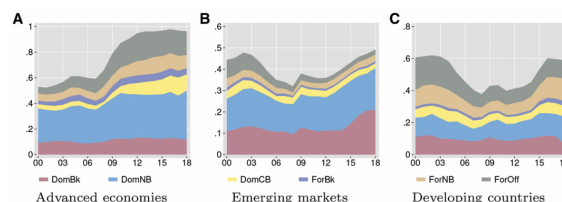


Figure 2
Sovereign debt holdings by investor group
The panels plot the aggregate debt by country group detailed in [Internet Appendix A](#) divided by the aggregate GDP of the country group. The investor group holdings are broken down into Domestic Banks (DomBK), Domestic Nonbanks (DomNB), Domestic Central Bank (DomCB), Foreign Banks (ForBK), Foreign NonBanks (ForNB), and Foreign Official (ForOff). Each panel shows a balanced sample over the given time frame.

5.2 Figure 2: Sovereign debt holdings by investor group

Read:

- The composition of creditor groups differs sharply across AEs, EMs, and DCs.
- AEs have growing official and central-bank involvement.
- EMs rely more on domestic investors overall, but foreign nonbanks still matter substantially.
- DCs show a larger role for foreign official creditors.

5.3 Table 1: Marginal holders of sovereign debt

Read:

- Across *all* countries, domestic investors absorb 65% of new debt and foreign investors 35%.
- Nonbanks are the key marginal investors: domestic nonbanks absorb 39% and foreign nonbanks 23% of new debt, jointly 62% of the increase.
- This is greater than proportional to their average holdings: domestic plus foreign nonbanks hold only 44% on average.
- In EMs, domestic nonbanks absorb 38% and foreign nonbanks 21%; foreign banks absorb only 6%.
- In DCs, foreign official creditors become more important on the margin.

5.4 Table 2: Marginal investors for U.S. Treasuries and U.K. Gilts

Read:

- Domestic nonbanks are again the main marginal absorbers of U.S. Treasuries and U.K. Gilts.
- In the U.S., domestic nonbanks absorb about 0.56 of each extra dollar issued.
- Within U.S. domestic nonbanks, money market funds (0.22) and household/hedge-fund type investors (0.16) account for roughly two-thirds of the nonbank response.
- In the U.K., domestic nonbanks absorb about 0.44 of marginal issuance, split mainly between funds/other financial firms and insurance/pension investors, with I&P especially important.

Table 1

Marginal holders of sovereign debt

	(1) Dom	(2) For	(3) DomBK	(4) DomNB	(5) DomCB	(6) ForBK	(7) ForNB	(8) ForOff
<i>A. Marginal share</i>								
All	0.65*** (0.03)	0.35*** (0.03)	0.20*** (0.03)	0.39*** (0.05)	0.07*** (0.02)	0.05*** (0.01)	0.23*** (0.03)	0.06*** (0.02)
AE	0.54*** (0.11)	0.46*** (0.11)	0.08* (0.04)	0.36*** (0.08)	0.09 (0.07)	0.06*** (0.02)	0.36*** (0.10)	0.05* (0.03)
EM	0.70*** (0.04)	0.30*** (0.04)	0.26*** (0.04)	0.38*** (0.07)	0.06** (0.02)	0.06*** (0.02)	0.21*** (0.04)	0.03 (0.02)
DC	0.61*** (0.06)	0.39*** (0.06)	0.10*** (0.03)	0.42*** (0.06)	0.09* (0.04)	0.03* (0.02)	0.22*** (0.07)	0.13*** (0.03)
<i>B. Average share</i>								
All	0.56	0.44	0.22	0.27	0.07	0.06	0.17	0.20
AE	0.56	0.44	0.21	0.30	0.05	0.10	0.21	0.13
EM	0.62	0.38	0.28	0.27	0.07	0.06	0.16	0.16
DC	0.48	0.52	0.14	0.24	0.09	0.04	0.14	0.34

Panel A reports the regression coefficients for Equation (2) for each investor group based on country groupings: All, Advanced Economies (AE), Emerging Markets (EM), and Developing Countries (DC). The first two columns represent domestic and foreign investors, respectively. Columns 3 through 8 correspond to the six investor groups. Standard errors clustered at the country level are reported in parentheses. The number of observations for each regression row are: All = 1947; AE = 438; EM = 927; DC = 582. Panel B reports the average share of debt held by each investor group. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 2
Marginal investors for U.S. and U.K. government debt

<i>A. U.S. Treasuries</i>									
	(1) Foreign	(2) DomCB	(3) DomBK	(4) DomNB	(5) MMFs	(6) HF/HH	(7) I&P	(8) O&F	(9) NonFin
UST	0.247*** (0.0389)	0.132*** (0.0395)	0.0606*** (0.0118)	0.560*** (0.0496)	0.212*** (0.0251)	0.162*** (0.0211)	0.109*** (0.0254)	0.0836*** (0.0235)	-0.0135 (0.0455)
Share of DomNB marginal Obs	104	104	104	104	104	104	104	104	104
<i>B. U.K. gilts</i>									
	(1) Foreign	(2) DomCB	(3) DomBK	(4) DomNB	(5) Funds&Oth	(6) I&P	(7)		
UKG	0.174*** (0.0290)	0.256*** (0.0473)	0.128*** (0.0251)	0.444*** (0.0470)	0.217*** (0.0434)	0.220*** (0.0413)	0.00996** (0.00384)		
Share of DomNB marginal Obs	104	104	104	104	104	104	104		

This table shows results from estimating Equation (3) for the holders of U.S. Treasuries (UST) and U.K. Gilts (UKG) over 1990Q1–2020Q4. For panel A, column 6 includes hedge funds, private equity, private trusts, and direct household holdings. Column 8 is all other financial institutions apart from funds and Insurance and Pension (I&P). Column 9 is all nonfinancial holders excluding households. For panel B, column 5 includes all financial firms besides banks, insurance companies, and pension funds. Column 7 is all nonfinancial holders. The rows labeled "Share of DomNB Marginal" provide the ratio of the coefficient estimates for each domestic nonbank investment group over the coefficient of total domestic nonbank holding in column 4. Holdings in panel A (B) are measured in U.S. dollars (British pounds). * $p < .1$, ** $p < .05$, *** $p < .01$.

5.5 Table 3: Euro-area holders of sovereign debt

Read:

- Euro-area investors account for a meaningful portion of foreign nonbank holdings of non-euro-area sovereign debt.
- Within euro-area nonbanks, the “other financial institutions” category—primarily investment funds—explains most of the marginal response.
- This group accounts for 78% of euro-area nonbank marginal investment overall and 89% for advanced-economy sovereigns outside the euro area.
- Insurance and pensions matter more for emerging-market destinations but remain secondary to investment funds.

Table 3

Euro area holders of sovereign debt

<i>A. Marginal investment from euro area (EA) investors</i>								
	(1) DomBK	(2) DomCB	(3) DomNB	(4) ForBK Non-EA	(5) ForBK EA	(6) ForNB Non-EA	(7) ForNB EA	(8) ForOff
Non-EA	0.12* (0.06)	0.06 (0.05)	0.62*** (0.12)	0.04 (0.03)	0.12* (0.01)	0.05* (0.06)	0.05* (0.02)	-0.01 (0.02)
Obs	124	124	124	124	124	124	124	124
R ²	0.592	0.349	0.653	0.228	0.267	0.321	0.577	0.261
<i>B. Euro nonbank decomposition, quarterly</i>								
	(1) Insur&Pens	(2) OtherFin	(3) Households	(4) NFC				
Non-EA	0.21* (0.12)	0.78*** (0.12)	0.01 (0.00)	0.01 (0.00)				
Obs	691	691	691	691				
R ²	0.323	0.826	0.084	0.058				
Non-EA EA	0.09 (0.06)	0.89*** (0.07)	0.00 (0.00)	0.01 (0.02)				
Obs	241	241	241	241				
R ²	0.428	0.927	0.214	0.123				
Non-EA EM	0.25 (0.14)	0.74*** (0.14)	0.01 (0.00)	0.00* (0.00)				
Obs	450	450	450	450				
R ²	0.345	0.803	0.086	0.118				

This table reports results for marginal investment in Non-EA sovereign debt by EA investors. Panel A estimates Equation (2) for 8 categories of investors by separating Foreign banks and nonbanks into EA (columns 5 and 7) and non-EA (columns 6 and 8). Panel B reports estimates from Equation (4) on sovereign debt issued by non-European countries. Standard errors clustered at country level are reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table 4

Investor demand: Emerging markets

	(1) DomBK	(2) DomNB	(3) DomCB	(4) ForBK	(5) ForNB	(6) ForOff
<i>A. IV estimates</i>						
Return - Domestic	4.913* (2.921)	9.898** (4.024)	13.205 (8.620)			
Return - Foreign				6.764* (3.661)	12.846** (5.902)	7.565* (4.430)
GDP	0.795*** (0.159)	1.111*** (0.277)	0.333 (0.494)	1.673*** (0.622)	2.586 (1.593)	0.523 (0.752)
GDP growth	3.849 (2.451)	6.721* (3.558)	12.174 (7.658)	3.230 (4.253)	18.992** (7.453)	8.362 (5.113)
Inflation	-3.820*** (1.118)	-4.075** (1.629)	-5.766* (3.160)	-6.730*** (2.400)	-7.433* (3.825)	-5.758** (2.885)
Exp-to-GDP	-0.248 (0.187)	-0.400 (0.300)	-0.785 (0.611)	-0.448 (0.340)	0.721 (0.837)	-0.332 (0.376)
Ratings	0.016 (0.013)	-0.002 (0.021)	0.051 (0.038)	-0.021 (0.026)	0.013 (0.044)	0.033 (0.029)
<i>B. Shares and elasticities</i>						
Mean $\alpha_j(n)$	0.140	0.340	NA	0.01	0.015	NA
Predictive coefficient ϕ_j	-0.57	-0.57	NA	-0.77	-0.77	NA
Price elasticity η_j	-2.41	-3.72	NA	-5.16	-9.74	NA
Yield elasticity $-T\eta_j$	12.04	18.62	NA	25.78	48.71	NA
Growth elasticity $\psi_{i,k}$	3.31	4.44	NA	3.20	18.71	NA
Inflation elasticity $\psi_{i,k}$	-3.29	-2.69	NA	-6.66	-7.32	NA

Panel A reports the EM sovereign debt IV estimates of the demand function in Equation (7), including country and year fixed effects. The sample spans 1990–2018 at annual frequency. The dependent variable is log holding of the group indicated in the column title. The standard errors clustered at country and year levels are reported in parentheses. Panel B gives the average portfolio share of sovereign debt, predictive coefficients and the implied price and yield demand elasticities, as well as demand elasticities with respect to real GDP growth and inflation. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.6 Table 4: Investor demand for EM sovereign debt

Read:

- Domestic nonbanks respond more strongly to expected excess returns than domestic banks.

- Foreign nonbanks are the most elastic EM investors, with implied yield elasticity around 48.7, compared with about 25.8 for foreign banks, 18.6 for domestic nonbanks, and 12.0 for domestic banks.
- Higher inflation reduces holdings across investor groups.
- Higher GDP growth generally raises holdings, especially for foreign nonbanks.

5.7 Table 5: Cost of financing sensitivity and counterfactuals

Read:

- For the average EM borrower, a 10% increase in debt raises yields by about 5.8% under the marginal-share calculation.
- Without nonbanks, the same debt increase raises yields by about 8.4%, showing that nonbanks materially stabilize sovereign funding costs.
- A 1 percentage point fall in real GDP growth raises yields by about 2.8% of the average EM yield.
- A 1 percentage point increase in inflation raises yields by about 1.69% of the average EM yield.
- Excluding nonbanks makes inflation sensitivity worse, while excluding banks makes GDP-growth sensitivity worse.

Table 5
Cost of financing sensitivity and counterfactuals

A. Cost of financing sensitivity to debt increase				
	Using average shares		Using marginal shares	
	Sensitivity	% Yield change	Sensitivity	% Yield change
Actual	0.43	4.9%	0.51	5.8%
No bank	0.34	3.8%	0.41	4.6%
No nonbank	0.69	7.9%	0.74	8.4%
B. Cost of financing sensitivity to characteristics				
	Real GDP growth decrease		Inflation increase	
	Sensitivity	% Yield change	Sensitivity	% Yield change
Actual	0.25	2.80%	0.15	1.69%
No bank	0.28	3.18%	0.13	1.44%
No nonbank	0.17	1.92%	0.20	2.27%
C. Cost of financing sensitivity to latent demand				
	DomBK	DomNB	ForBK	ForNB
Sensitivity	-1.28	-1.23	-0.32	-0.73
D. Components from prior table estimates				
	DomBK	DomNB	ForBK	ForNB
Price elasticity η_i	-2.41	-3.72	-5.16	-9.74
Average shares θ_i	0.36	0.35	0.08	0.21
Marginal shares a_i	0.29	0.42	0.07	0.23
GDP growth $\psi_{i,k}$	3.31	4.44	3.20	18.71
Inflation $\psi_{i,k}$	-3.29	-2.69	-6.66	-7.32

Panel A reports the cost of financing sensitivity to debt increase in Equation (15) and (16) based on, alternatively, average portfolio shares and marginal responses. Rows are based upon the “Actual” for all investors, for “No bank” and “No nonbank” in turn assuming shares are redistributed to other investors. Panel B gives borrowing cost sensitivity due to the characteristics as in Equation (18). Panel C provides the effects of latent demand by investor group given in Equation (19). Panel D provides the inputs used in the counterfactual calculations. The marginal shares a_i and average shares θ_i are from Table 1 and rescaled. The price elasticities η_i , and characteristics elasticities ψ_i for GDP growth and inflation are from Table 4. All percentage change calculations are based on average EM 5-year yield in our sample of 8.8%.

Table 6
Advanced economy sovereign debt holdings

	(1) DomBK	(2) DomNB	(3) DomCB	(4) ForBK	(5) ForNB	(6) ForCB
Bond yield	1.196*** (0.333)	0.953*** (0.304)	2.742*** (0.774)	0.885*** (0.290)	2.819*** (1.061)	0.689*** (0.186)
GDP growth	7.785** (3.460)	-0.184 (3.098)	19.948** (8.777)	2.683 (3.039)	14.968* (8.747)	4.899** (2.324)
GDP	-0.052 (0.488)	0.294 (0.433)	0.397 (1.301)	2.050*** (0.427)	-0.323 (1.322)	1.972*** (0.366)
Exp-to-GDP	1.274* (0.696)	0.718 (0.595)	8.167*** (1.831)	2.096*** (0.610)	6.066** (2.471)	1.352** (0.584)
Ratings	-0.139*** (0.042)	-0.091** (0.037)	-0.308*** (0.104)	-0.118*** (0.037)	-1.455** (0.604)	0.7 (0.026)
Inflation	5.142 (4.574)	3.846 (3.858)	30.277** (12.074)	14.965*** (4.017)	6.764 (9.946)	10.857*** (3.990)
Observations	316	298	290	317	305	291

This table reports the estimates of the Equation (20). The sample spans 2000–2018 at an annual frequency. The dependent variable is holding of the group indicated in the column title. The standard errors clustered at country and year levels are reported in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.8 Table 6: Advanced-economy sovereign debt holdings

Read:

- In the AE robustness exercise, all investor groups’ holdings rise with yields.

- Foreign nonbanks are again the most responsive group, with a yield coefficient of about 2.82 versus 0.89 for foreign banks.
- The pattern echoes the EM results, even though the AE exercise is more limited because the full structural assumptions do not hold there.

6 Short summary

- **Conclusion.** Sovereign debt composition matters because investor groups have different demand elasticities.
- **Main identification insight.** The paper uses the fact that the full sovereign debt stock is allocated across investor groups to build an issuer-side market-clearing instrument for debt demand.
- **Main empirical lesson.** Nonbank private investors—especially investment-fund-type investors—are the key marginal absorbers of sovereign debt and the most elastic EM creditors.
- **Main policy lesson.** EM sovereigns are especially exposed to losing nonbank investors because such losses sharply raise the borrowing-cost response to new debt issuance.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that sovereign borrowing conditions depend on investor composition, not just total demand.
- It documents that nonbank private investors absorb a disproportionate share of sovereign debt expansions.
- It then shows structurally that these same investors are the most price elastic, especially in emerging markets.

7.2 Economic intuition

- If highly elastic investors hold a large share of debt, the sovereign can place additional debt with smaller yield increases.
- If instead the debt is held by less elastic investors, the same debt increase requires a larger price concession.
- Nonbanks matter precisely because they do both: they hold a large amount on the margin and they are comparatively willing to adjust portfolios when yields move.

7.3 Takeaway

This paper is best understood as a sovereign-debt application of the modern demand-system approach to asset pricing. Its main contribution is to show that *who* holds sovereign debt is not a descriptive

curiosity but a pricing object. In the data, nonbanks—especially investment-fund-type investors—are the dominant marginal absorbers of debt expansions. In the structural estimates, they are also the most elastic creditors. The result is a sharp policy implication: emerging-market sovereigns are especially vulnerable to the loss of nonbank investors because that loss meaningfully increases the financing cost of additional borrowing.